Case 01: Supersonic Double Wedge Airfoil

Case Details

The objective of the simulation is to validate the experimental data given in [1] with low computational cost for fast but accurate results.

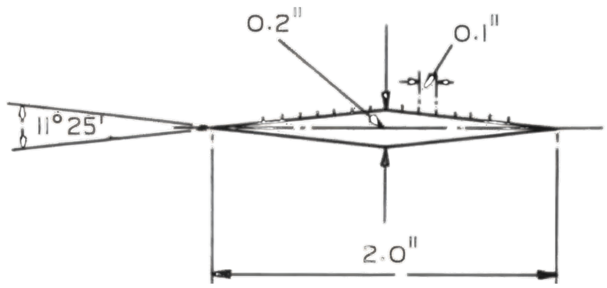

Figure 1: Double wedge airfoil as given in [1]

- Reference: https://reports.aerade.cranfield.ac.uk/handle/1826.2/3352
- Airfoil: Double wedge airfoil with half angle of $\approx 5°$
- Mach Number: 1.86
- Reynolds Number: 3.51×10^7
- Angle of Attack: $2°$

How to Run

1. Select wedge airfoil from the airfoil directory
2. Enter the desired number of coordinates if required and output directory.
3. Click on "Analyze" to obtain the accurate parameterization method and its respective parameters
4. Go to SU2 Settings → Meshing & Conditions

5. Enter Mach, Reynolds number and enter 4 in minimum AoA with 0 step.
6. Select "Load recommended settings"
7. Choose the type of mesh and target y+ value
8. Select "Generate Hybrid Mesh"
9. Go to SU2 Settings → Solver Setup & Run
10. Modify settings as desired or select "Run SU2 Analysis" (validated settings have been provided as default and are given here in the next section)
11. After convergence, the output or post-processed contours will be displayed in the SU2 Settings → Results tab and saved in the output directory.

Solver Settings

Given below are the solver settings validated for this case and are provided by default for this regime:

Setting	**File (supersonic.cfg)**
NUM_METHOD_GRAD	WEIGHTED_LEAST_SQUARES
CONV_NUM_METHOD_FLOW	ROE
SLOPE_LIMITER_FLOW	VENKATAKRISHNAN
LINEAR_SOLVER	FGMRES
LINEAR_SOLVER_PREC	ILU
KIND_TURB_MODEL	SST (V1994m)
CFL_NUMBER	0.5
CONV_FIELD	RMS_DENSITY, RMS_TKE
Min log10 Residual	-6

Table 1: Settings used in `supersonic.cfg`

Result

AoA	Cl_{exp}	Cd_{exp}	Cl_{obt}	Cd_{obt}	**Cl Error (%)**	**Cd Error (%)**
2	0.082424	0.023529	0.087857	0.026282	6.591527	11.70045

Table 2: Results at Mach 1.86 with comparison between experimental and obtained data

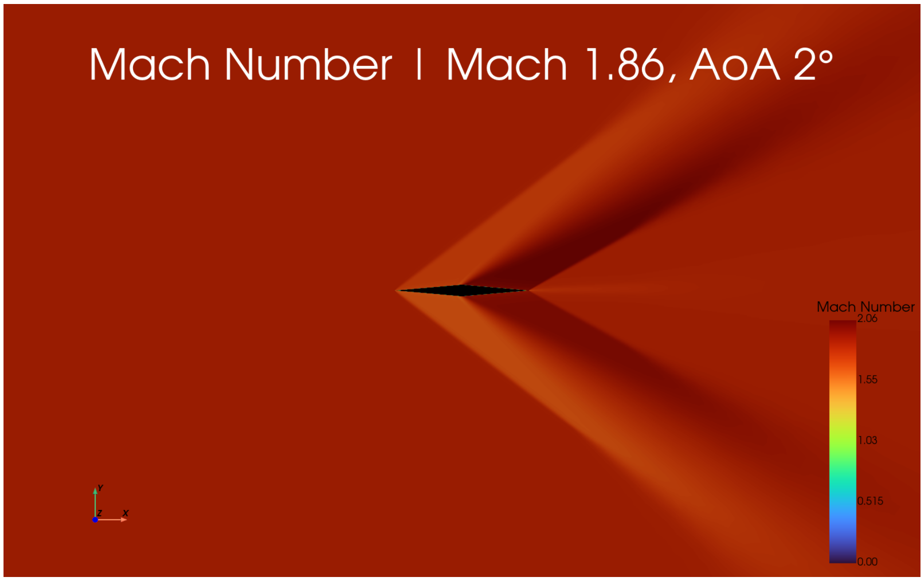

Figure 2: Mach number field

Figure 3: Pressure field

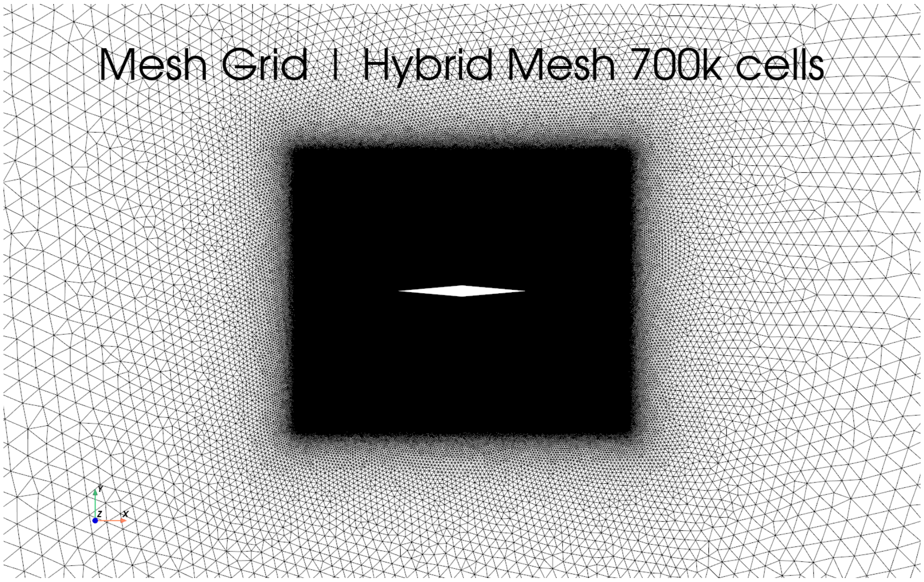

Figure 4: Mesh visualization

References

[1] D. Beastall and R. J. Pallant, *Wind-tunnel tests on two-dimensional supersonic aerofoils at* $M = 1.86$ *and* $M = 2.48$, Aeronautical Research Council Reports and Memoranda No. 2800, 1950.